

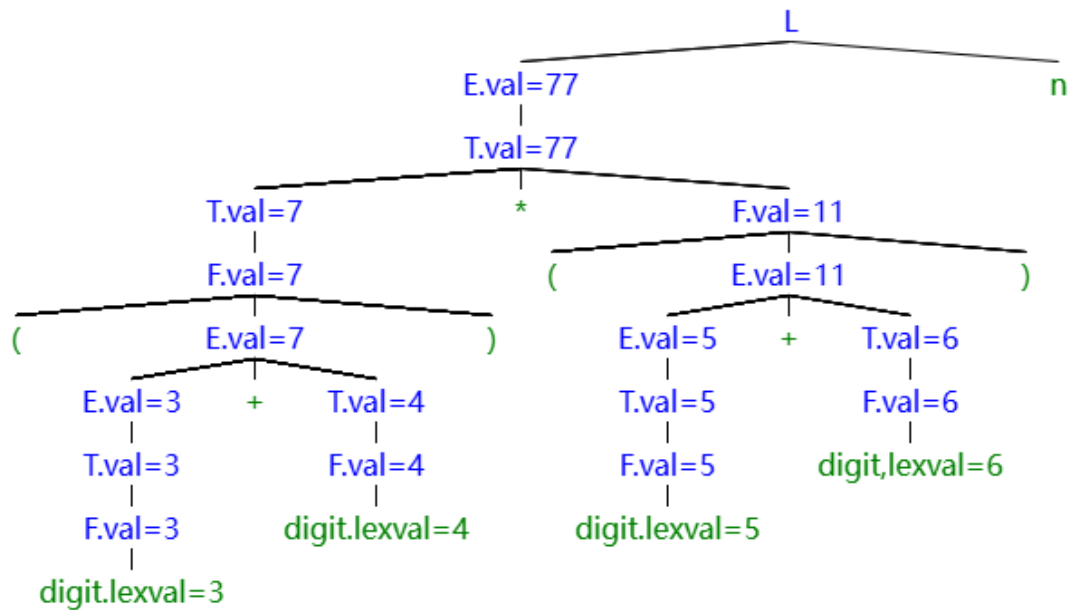
Assignment 5

Name: 项枫 Student ID: 2022211570 Class Number: 2022211801

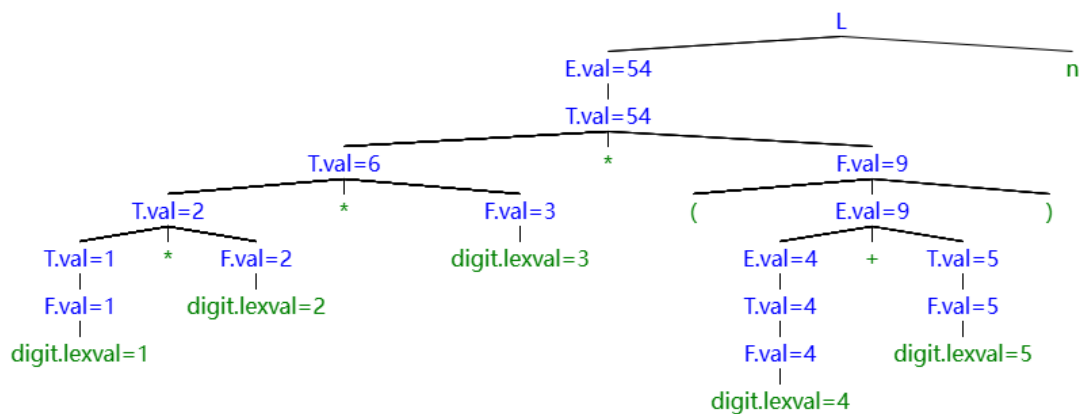
1 Required Exercises

Exercise 1

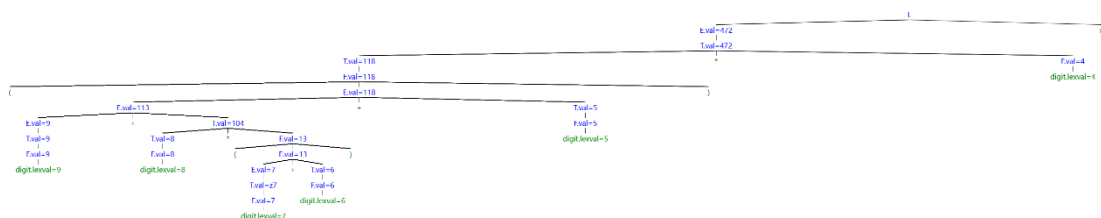
1、 $(3 + 4) * (5 + 6)n$

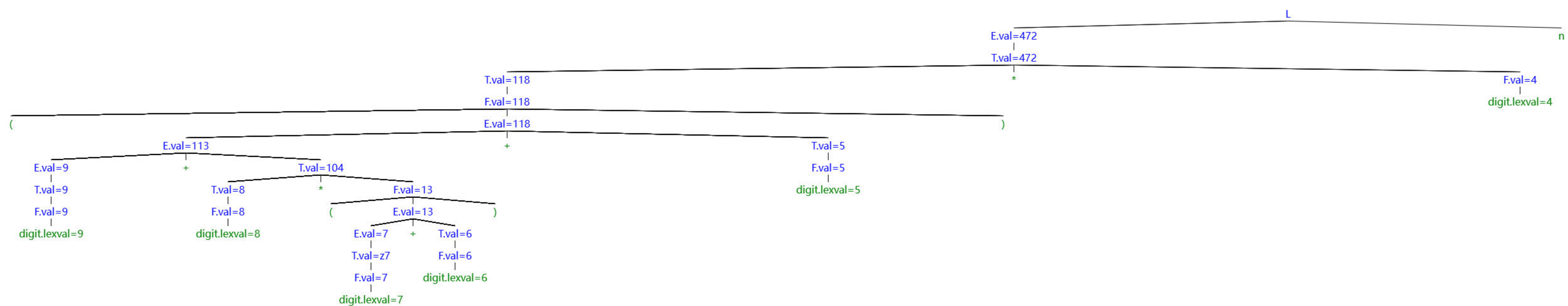


2、 $1 * 2 * 3 * (4 + 5)n$



3、 $(9 + 8 * (7 + 6) + 5) * 4n$





Exercise 2

There are totally 10 different topological sorts for the dependency graph.

- 1: 1, 2, 3, 4, 5, 6, 7, 8, 9;
- 2: 1, 2, 3, 5, 4, 6, 7, 8, 9;
- 3: 1, 2, 4, 3, 5, 6, 7, 8, 9;
- 4: 1, 3, 2, 4, 5, 6, 7, 8, 9;
- 5: 1, 3, 2, 5, 4, 6, 7, 8, 9;
- 6: 1, 3, 5, 2, 4, 6, 7, 8, 9;
- 7: 2, 1, 3, 4, 5, 6, 7, 8, 9;
- 8: 2, 1, 3, 5, 4, 6, 7, 8, 9;
- 9: 2, 1, 4, 3, 5, 6, 7, 8, 9;
- 10: 2, 4, 1, 3, 5, 6, 7, 8, 9.

Exercise 3

PRODUCTIONS	SEMANTIC RULES
1) $E \rightarrow E1 + T$	$E.type := float \text{ if } E1.type = float \text{ else } int$
2) $E \rightarrow T$	$E.type := T.type$
3) $T \rightarrow num.num$	$T.type := float$
4) $T \rightarrow num$	$T.type := int$